N O N - N O R M A T I V E

`T = sigma_v    det = -1    (x, y) -> (-x, y)`

"This reflects the standard."

Informative / Advisory (NOTE)

Owl Semaphore — Non-Normative

OWL 2 / Reflection State (sigma_v) / Standard Specification

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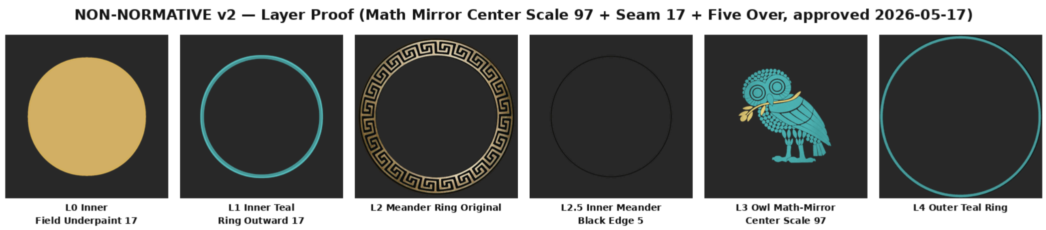
SOURCE github.com/IT-Help-San-Diego/owl-semaphore VERSION v2.0.0-rc · LICENSE CC-BY-4.0

BANNER-TUPLE :: STATE=NON-NORMATIVE :: TRANSFORM=T = sigma_v   det = -1   (x, y) -> (-x, y) :: QUOTE="This reflects the standard." :: VERSION=v2.0.0-rc :: CONCEPT-DOI=10.5281/zenodo.19473697 :: PUBLISHED-VERSION-DOI=10.5281/zenodo.19474599 :: RC-VERSION-DOI=TBD_BY_ZENODO_ON_RELEASE

NON-NORMATIVE — LAYER PROOF PALETTE



Presentation-layer composed badges: per-state palette meander/ring around the V4-tested owl-only master. The mathematical master is the owl-only transparent PNG (shown below); the surrounding circle is editorial.

OWL-ONLY MATHEMATICAL MASTER — V4-TESTED SOURCE

These are the V4-tested owl-only PNGs (owl + human-selected gold branch). They are the algebraic master; tests/test_v2_assets.py gates them. The surrounding circle/meander above is presentation only and is not part of the V4 transform input.

1. Da Vinci's Wings

State line: NON-NORMATIVE — $T = \sigma_v$, det $= -1$, $(x, y) \mapsto (-x, y)$

Stand in front of a mirror. You are still upright — up is still up, down is still down — but left and right have swapped. That is a vertical-axis reflection: $(x, y) \mapsto (-x, y)$. Not a rotation — a mirror image.

When Leonardo da Vinci spent years studying birds, sketching wing mechanics, and building machines that could not fly, he was a Non-Normative owl. Facing the other direction. Seeing what everyone else had their backs to. He failed, but he left the rest of the world a guideline for success. The Wright Brothers stood on his shoulders 400 years later. Non-normative work is the engine of progress: rigorous exploration that hasn't finished yet.

> This section is **orientation, not proof.** The formal scientific object remains the reflection operator σ_v, with determinant -1 and coordinate action $(x, y) \mapsto (-x, y)$. The story above is the plain-English meaning of that operator; the operator itself is defined formally in §5 (Mathematical Definition).

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2. Statement of Intent

This document defines the **NON-NORMATIVE owl** as a formal epistemic state within the Owl Semaphore system.

This is not a decorative variation of the normative mark. It is a mathematically defined reflection state with a specific role in analysis, interpretation, and structured disagreement.

The goal is to preserve rigor while allowing controlled deviation from the baseline framework.

3. System Context

The Owl Semaphore system is defined by the Klein four-group:

$$V_4 = \{I, \sigma_v, C_2, \sigma_h\}$$

The NON-NORMATIVE owl corresponds to the vertical reflection operator:

$$\sigma_v : (x, y) \mapsto (-x, y)$$

4. Ontological Role

4.1 Semantic Designation

NON-NORMATIVE represents:

- legitimate alternative interpretation
- reflective critique
- analytical deviation
- structured disagreement

4.2 Interpretive Role

This state indicates that the content:

- departs from the canonical model
- remains structurally grounded in the same system
- is not arbitrary or invalid

4.3 What It Does Not Mean

- not random
- not incorrect by default
- not adversarial (that is CRITICAL)

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It is **reflection**, not rejection.

5. Mathematical Definition

5.1 State Operator

$$T_{\text{non}} = \sigma_v$$

5.2 Matrix Form

$$\sigma_v = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

5.3 Determinant

$$\det(\sigma_v) = -1$$

5.4 Properties

- orientation-reversing
- reflection class
- order 2 (self-inverse involution)

$$\sigma_v \circ \sigma_v = I$$

6. Coordinate System

The NON-NORMATIVE state uses the same coordinate system as NORMATIVE:

- canvas: 1080×1080
- center: (540, 540)

Transformation is applied relative to this center.

7. Canonical Orientation

7.1 Visual Definition

- upright
- faces LEFT

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7.2 Transform Relationship

The NON-NORMATIVE owl is the horizontal mirror of the normative owl.

8. Asset Topology

The approved Math-Mirror Center-Scale-97 master extends the four-layer NORMATIVE topology with two seam-refinement layers:

- L0 — inner field underpaint (17 px)
- L1 — inner teal ring outward (17 px)
- L2 — meander ring (original, shared with NORMATIVE)
- L2.5 — inner meander black edge (5 px over)
- L3 — owl body (math-mirror center-scale-97)
- L4 — outer teal ring

L2.5 and the 17-px seam treatments in L0/L1 are presentation-layer refinements to maintain legibility on the dark inner field; they do not enter the V_4 transform. The mathematical state operator remains σ_v.

8.1 Composite Definition

$$N_{\text{non}} = L_0 \oplus L_1 \oplus L_2 \oplus L_{2.5} \oplus L_3 \oplus L_4$$

9. Geometry

All geometric constraints are inherited from the normative standard:

- identical radii
- identical center
- identical annular structure

No geometric deformation is permitted.

10. Color Specification

10.1 Palette

The approved Math-Mirror Center-Scale-97 + Seam-17 + Five-Over master uses the following palette on the dark inner field:

- owl: teal — observed dominant RGB **(77, 177, 176)** ≈ `#4DB1B0`

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- outer ring: teal — same family as the owl, recolored from the geometry layer
- meander: gold (unchanged, shared with NORMATIVE)
- inner field: dark slate (carried over from the L0 inner-field underpaint)

The full reviewed provenance for these layers (L0 inner-field underpaint 17, L1 inner teal ring outward 17, L2 meander ring original, L2.5 inner meander black edge 5-over, L3 owl math-mirror center-scale-97, L4 outer teal ring) lives in `assets/v2/nonnormative-math97-five-over-master/`.

10.2 Color Doctrine

Teal represents analytical distance from the normative baseline while maintaining structural coherence. The brighter teal used in the approved Math-Mirror Center-Scale-97 master is calibrated for legibility against the dark inner field, mirroring the parchment-tone rationale used for NORMATIVE.

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11. Transparency and Alpha

Same rules as normative:

- RGBA required
- corner alpha = 0
- center alpha = 255

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12. Provenance

12.1 Construction

The approved master is the human-reviewed **Math-Mirror Center-Scale-97 + Seam-17 + Five-Over** composite, staged in `assets/v2/nonnormative-math97-five-over-master/`. Its layer order is:

1. L0 — inner field underpaint 17
2. L1 — inner teal ring outward 17
3. L2 — meander ring original (shared geometry with NORMATIVE)
4. L2.5 — inner meander black edge 5 over (seam refinement)
5. L3 — owl math-mirror center-scale-97 (vertical-axis mirror of the source owl, then re-centered and scaled to 97 %)
6. L4 — outer teal ring

The owl-only L3 is the visual mathematical master for NON-NORMATIVE. It is *close to* but not bit-for-bit identical with σ_v`(NORMATIVE)` because of the 97 % re-scale and the seam refinements introduced for legibility on the dark inner field. The asset was promoted only after explicit human visual approval; the source TIFF, all six layers, and the audit JSON are preserved verbatim under the master kit directory for future review.

12.2 Transform Integrity

The state's *formal* operator remains σ_v with determinant -1. The visual master is a presentation-layer asset built from σ_v-derived owl geometry plus reviewed seam refinements; the algebra of the state system is unchanged. The full audit trail (per-layer SHA-256, diff-bbox check against the user-approved proof) lives in `assets/v2/nonnormative-math97-five-over-master/metrics/OWL-2-NON-NORMATIVE-MATH97-FIVE-OVER-METRICS.json`.

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13. Asset Invariants

13.1 Algebraic

- operator $= \sigma_v$
- determinant $= -1$

13.2 Visual

- upright
- mirrored orientation

13.3 Structural

- geometry unchanged
- layer structure unchanged

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14. Integrity Regime

All assets must:

- pass SHA-3-512 verification
- be reproducible from layers

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15. Interpretation Rules

15.1 Positive Rule

When present:

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The content reflects the normative framework while offering a structured alternative.

15.2 Negative Rule

It does not indicate failure or error.

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16. Non-Permitted Changes

- rotation
- vertical flip
- C2 inversion
- geometry alteration

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17. Relationship to Other States

- NORMATIVE → identity (*"This is the standard."*)
- NON-NORMATIVE → reflection (*"This reflects the standard."*)
- CRITICAL → inversion (*"This inverts the standard."*)
- METACOGNITIVE → frame-audit (*"The observer audits the frame."* — thinking examines its own frame)

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18. Formal Definition

Let L_1–L_4 be the layer fields.

$$N_{\text{non}} = L_1 \oplus L_2 \oplus L_3 \oplus L_4$$

with

$$T = \sigma_v$$

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19. Closing Statement

The NON-NORMATIVE owl preserves system integrity while enabling structured deviation.

It is the mechanism by which the system permits disagreement without collapse.

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OWL SEMAPHORE SYSTEM — CLASSIFICATION LEDGER (v2.0.0-rc)

NORMATIVE

$T = I \quad \det = +1$

(x, y) -> (x, y)

"This is the standard."

RFC 2119 MUST / SHALL

NON-NORMATIVE

$T = \sigma_v \quad \det = -1$

(x, y) -> (-x, y)

"This reflects the standard."

Informative / Advisory (NOTE)

CRITICAL

$T = C_2 \quad \det = +1$

(x, y) -> (-x, -y)

"This inverts the standard."

RFC 2119 MUST NOT / SHALL NOT

METACOGNITIVE

$T = \sigma_h \quad \det = -1$

(x, y) -> (x, -y)

"The observer audits the frame."

Epistemic / Framework (META)

Accessibility: color is not the only carrier. State identity is recoverable from color + orientation + textual label (WCAG 2.2 SC 1.4.1).

Owl Semaphore v2.0.0-rc · github.com/IT-Help-San-Diego/owl-semaphore
Concept DOI 10.5281/zenodo.19473697 · v1.2.0 DOI 10.5281/zenodo.19474599 · v2.0.0 DOI TBD_BY_ZENODO_ON_RELEASE